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Nonlinear projection methods for visualizing Barcode data and application on two data sets

MADALINA OLTEANU,* VIOLAINE NICOLAS,† BRIGITTE SCHAEFFER,‡ CHRISTIANE DENYS,† ALAIN-DIDIER MISSOUP,§ JAN KENNIS¶ and CATHERINE LARÉDO‡**
*SAMM (Statistique, Analyse et Modélisation Multidisciplinaire), EA 4543, Université Paris 1 Panthéon Sorbonne, 90 rue de Tolbiac, Paris, 75013, France, †Département Systématique et Evolution, UMR CNRS 7205, Museum National d'Histoire Naturelle, 47 rue Cuvier, CP 51, Paris, 75005, France, ‡Mathématiques et Informatique Appliquées, INRA, UR341, Jouy-en-Josas, 78350, France, §Department of Animal Biology Organisms, Faculty of Science, The University of Douala, PO Box 24157, Douala, Cameroon, ¶Evolutionary Ecology Group, University of Antwerp, Groenenborgerlaan 171, Antwerpen, B-2020, Belgium, **LPMA, UMR CNRS 7599, Université Paris Diderot, Paris, 75013, France*

Abstract

Developing tools for visualizing DNA sequences is an important issue in the Barcoding context. Visualizing Barcode data can be put in a purely statistical context, unsupervised learning. Clustering methods combined with projection methods have two closely linked objectives, visualizing and finding structure in the data. Multidimensional scaling (MDS) and Self-organizing maps (SOM) are unsupervised statistical tools for data visualization. Both algorithms map data onto a lower dimensional manifold: MDS looks for a projection that best preserves pairwise distances while SOM preserves the topology of the data. Both algorithms were initially developed for Euclidean data and the conditions necessary to their good implementation were not satisfied for Barcode data. We developed a workflow consisting in four steps: collapse data into distinct sequences; compute a dissimilarity matrix; run a modified version of SOM for dissimilarity matrices to structure the data and reduce dimensionality; project the results using MDS. This methodology was applied to *Astraptes fulgerator* and *Hylomyscus*, an African rodent with debated taxonomy. We obtained very good results for both data sets. The results were robust against unbalanced species. All the species in *Astraptes* were well displayed in very distinct groups in the various visualizations, except for LOHAMP and FAB-OV that were mixed up. For *Hylomyscus*, our findings were consistent with known species, confirmed the existence of four unnamed taxa and suggested the existence of potentially new species.

Keywords: Barcode, dissimilarity matrices, DNA sequences, median self-organizing maps, multidimensional scaling, unsupervised algorithms, visualization

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Introduction

The scope of the DNA barcode project was to provide a standard system for fast and accurate identification initially of animal species and later on of all eukaryotic species based on a 648 bp region of the CO1 mitochondrial gene (Hebert *et al.* 2003; Miller 2007). DNA barcoding is now recognized as a valuable addition to the taxonomic toolbox. DNA barcoding aims first at assigning unknown specimens to described species and second at enhancing the discovery of new species and/or providing better descriptions of cryptic species (see e.g. Hebert *et al.* 2004; DeSalle *et al.* 2005; Frezal & Leblois 2008; Casiraghi *et al.* 2010; Taylor & Harris 2012). For studying the first goal, various methods have been proposed. Some methods rely on biological assumptions: phylogenetic methods (as neighbour-joining trees, Saitou & Nei 1987; or phyML, Guindon & Gascuel 2003); methods based on coalescent theory coupled with a maximum likelihood or Bayesian approach (Abdo & Golding 2007; Munch *et al.* 2008). Other methods do not rely on biological evolution models and develop algorithms looking for characteristics attributes (CAOS, Sarkar *et al.* 2002). Another approach is to put this query assignment problem in a purely statistical context so that it becomes a 'supervised classification problem' and all the various algorithms developed in this framework can be used: some are distance-based (k-NN, Fix & Hodges 1951), character-based (CART, Breiman *et al.* 1984; Random Forest, Breiman 2001), or based on more complex tools (Kernel and SVM

Correspondence: Catherine Larédo, Fax: 33 1 34 65 22 17;
E-mail: catherine.laredo@jouy.inra.fr

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methods, Shawe-Taylor & Cristianini 2004). A comparison of various phylogenetic and statistical methods in the context of DNA barcoding is given in Austerlitz *et al.* (2009).

DNA barcodes can also be used as an exploratory tool for studying species delimitation and species discovery. This is much more controversial and raises many questions about the molecular delimitation of species. Various methods have been proposed such as NJ trees, methods based on a 'barcoding gap' (Hebert *et al.* 2004; O'Meara 2010; Puillandre *et al.* 2012), methods based on population genetic models (Matz & Nielsen 2005; Pons *et al.* 2006; David *et al.* 2012), Bayesian phylogenetic approaches methods based on multilocus data (Yang & Rannala 2010). The reliability of these methods has also been studied (see e.g. Meyer & Paulay 2005; Hickerson *et al.* 2006; Ross *et al.* 2008). Although all these approaches add valuable information, 'species delimitation' based on DNA sequences remains complex and is far from being solved, even if any taxonomic decision based on Barcode data has to be independently confirmed. As for the assignment problem, species delimitation can also be put in a purely statistical context which is unsupervised classification or clustering, where, contrary to the previous framework, no classes are *a priori* defined. Clustering combined with projection methods has usually two objectives. The first one is visualization of data to identify groups or show affinities between samples by displaying similar samples close together. The second one is finding structure in the data, i.e. grouping a collection of objects into subsets or clusters such that those within each cluster are more closely related to one another than to objects assigned to different clusters.

In this article, we will not directly address this species identification issue, but instead attempt to derive a visualization algorithm for Barcode data by means of statistical learning, clustering being studied mainly through visualization. Except in Sirovich *et al.* (2009), visualizing Barcode data has been little studied in the literature. These authors studied visualization in terms of correlation matrices which differs from the viewpoint presented here. In this article, we used nonlinear-projection algorithms such as Multidimensional Scaling (MDS, Gower 1966) and Self-Organizing Maps (SOM, Kohonen & Somervuo 1998; Kohonen 2001a,b) to visualize data and also consider, at least partially, the problem of cryptic species and of species delimitation. Both MDS and SOM methods map data onto a lower dimensional manifold. However, they approach this issue in somewhat different ways. Roughly speaking, starting from N observations in a high dimensional space, MDS looks for N points on a low dimensional space that best preserve pairwise distances. Usually, the plane is chosen for its simplicity. The SOM algorithm looks for a projection on the plane that best preserves the topology of the data. Multidimensional Scaling is a visualization algorithm based on dissimilarity matrices that can be used for both categorical and quantitative data. However, for large data sets, it might be rapidly unreadable as the algorithm does not organize the data. Self-Organizing Maps, on the other hand, organize the data set while projecting it. Initially, SOM were designed for multidimensional real data for which the Euclidean distance is a natural distance between observations. They can be used both for visualizing data and for finding clusters in the data. SOM have already been used for analysing gene expression data (Toronen *et al.* 1999; Nikkila *et al.* 2002), which are multidimensional real data. However, to our knowledge, they have never been used for visualizing or clustering DNA sequences. The main reason is probably that, contrary to gene expression data, observations are no longer Euclidean as they consist of sequences of nucleotides, so that algorithms have to deal with categorical data instead of multidimensional vectors.

For studying Barcode data, we develop here a methodology which partly relies on a modified version of the SOM algorithm for dissimilarity matrices (Conan-Guez *et al.* 2006). The study is organized as follows. We first give a short recap on multidimensional scaling and self-organizing maps, then we present our methodology for Barcode data. It consists of four steps: first, the data are cleaned up and collapsed into distinct sequences; second, a dissimilarity matrix is computed using any genetic distance or dissimilarity; third, the SOM algorithm for dissimilarity matrices is run to structure the data set and reduce dimensionality, and finally data are projected using MDS. This methodology is then applied to two very different data sets, the CO1 *Astraptes fulgerator* data set (Hebert *et al.* 2004) and a more complex data set composed of Cytb DNA sequences of a very diverse African rodent, the *Hylomyscus* genus whose taxonomy is still debated.

Materials and methods

Both MDS and SOM methods, as most projection algorithms, were initially designed for data lying in a Euclidean space. Subsequent developments, however, can deal with dissimilarity measures. As DNA sequences are categorical data, there is no underlying natural Euclidean space and only measures of dissimilarity between sequences are available. For Barcode data, dissimilarities between sequences were evaluated using specific measures, such as the Kimura-2P distance or the Hamming distance (i.e. normalized number of differences, also called p-distance). Hence, starting from a data set X with N observations, the information contained in X was

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summed up in a $N \times N$ dissimilarity matrix $D = (d_{ij})_{i,j=1,\ldots,n}$ with $d_{ij} = d(x_i, x_j)$ the dissimilarity between observations x_i and x_j.

Multidimensional scaling (MDS)

Multidimensional scaling comprises a wide family of visualization methods. The aim of the algorithm is to represent data on a plane while preserving pairwise distances or dissimilarities between data. More precisely, MDS seeks N points z_i in $\mathbb{R}^2$ to minimize a stress function

$$E = \sum_{i<j} \left(\delta_{ij} - d_{ij}\right)^2,$$

where $\delta_{ij} = ||z_i - z_j||$ represents the usual distance between the projections z_i, z_j of x_i, x_j.

Throughout the article we use a variant of MDS called Sammon mapping or Nonlinear mapping (NLM, Sammon 1969), where the emphasis is to preserve small distances between data: the stress function to minimize becomes

$$E = \frac{1}{\sum_{i<j} d_{ij}} \sum_{i<j} \frac{\left(\delta_{ij} - d_{ij}\right)^2}{d_{ij}}.$$

Nonlinear mapping (NLM) is a powerful projection method that can handle nonlinear manifolds, at least if they are not too heavily folded. It is a simple and elegant algorithm which may be used as a first step in the analysis of a data set. As the NLM stress function is weighted to represent accurately small dissimilarities, it is well adapted for detecting clusters in a data set. Note that, because of the weighting, NLM does not accept zeros outside the diagonal of the dissimilarity matrix D.

Self-organizing maps for dissimilarity data (DSOM)

Self-organizing maps are among the most popular neural networks techniques. Their interest is double: samples are grouped into homogeneous cells represented by prototypes (generally the centroid), and projected onto a lower dimensional space (usually, two dimensions) endowed with a topological structure. Thus, SOM perform both a partition of the data and a nonlinear projection. Due to its construction, the algorithm has the nice property of preserving the topology of the data while projecting it, so that close samples are projected onto the same cell or neighbouring cells. Initially, SOM was designed for quantitative vector data for which computations of Euclidean distances and centroids are immediate. Here, we used a variation of SOM (noted DSOM hereafter) for dissimilarity matrices proposed by Conan-Guez *et al.* (2006) detailed below in the special case of a SOM defined on a finite two-dimensional rectangular grid.

First, an *a priori* structure described by an undirected graph $\mathcal{G} = (\mathcal{M}, E)$ is given, where $\mathcal{M}$ is the set of nodes with cardinality M and E the set of edges. The SOM procedure tries to bend the plane so that the M nodes (named prototypes hereafter) *a priori* organized sum up the data as well as possible. The distance between two nodes of G, denoted $h(k, l)$, is the shortest path between these nodes. The algorithm associates to each node k a subset A_k of the data set X and a prototype $m_k \in A_k$ such that these M subsets form a disjoint partition of X. Each observation x_i has a class defined by $c(x_i) = k$ if x_i belongs to A_k. The prototypes m_k are usually chosen as the medoids of A_k (i.e. the point of A_k that minimizes the sum of dissimilarities in A_k). To get an organization of the map along the iterations, a neighbourhood function K indexed by a parameter T decreasing with time is introduced, $K_T(u) = K(u) = \exp\left(-\frac{u^2}{2T}\right)$. The aim of the algorithm is to find a partition $\mathcal{A} = (A_k)$ of X into disjoint classes and a set $\mathcal{M} = (m_1, \ldots, m_M)$ of prototypes that minimize the extended within-class variance,

$$E(\mathcal{A}, \mathcal{M}) = \sum_{i=1}^{N} \sum_{k=1}^{M} K(h(c(x_i), k))\ d(x_i, m_k)$$

$c(x_i)$ is the class of x_i $h(.,.)$ the graph distance, $d(.,.)$ the dissimilarity measure on X.

In the version proposed by Conan-Guez *et al.* (2006), the algorithm proceeds iteratively in two successive steps, the assignation step and the representation step. First, the prototype set $\mathcal{M}_0 = (m_{0,1}, \ldots, m_{0,M})$ is initialized at random among the samples. From the prototypes set, the l-th iteration $\mathcal{M}_{l-1}$ performs as follows:

The assignment step. During this step, the prototypes are fixed to $\mathcal{M}_{l-1}$. For each data point x_i and each prototype $m_{l-1,k}$, the algorithm computes

$$\gamma\left(x_i, m_{l-1,k}\right) = \sum_{k'=1}^{M} K(h(k', k))\ d\left(x_i, m_{l-1,k'}\right).$$

Then, the new partition is obtained actualizing the class of x_i according to the equation,

$$c_l(x_i) = \mathrm{Argmin}_{k \in \{1,\ldots,M\}} \gamma\left(x_i, m_{l-1,k}\right).$$

The representation step. During this step, the partition $\mathcal{A}_l$ is fixed and the M prototypes are updated as follows: for each node k, the algorithm computes successively

$$E_k(x_j) = \sum_{i=1}^{N} K(h(c(x_j), k))\ d(x_i, x_j) \text{ and}$$

$$m_{l,k} = \mathrm{Argmin}_{j \in \{1,\ldots,N\}} E_k(x_j).$$

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Then, a new set of prototypes $\mathcal{M}_l = (m_{l,1}, \ldots, m_{l,M})$ is obtained. The algorithm is stopped when a maximum number of iterations is achieved or when the extended within-class variance does not improve any longer. The convergence is achieved rather fast, usually in less than 10 iterations. Let us remark that DSOM is a BATCH algorithm (each iteration uses all samples).

According to Fort *et al.* (2002), BATCH SOM for Euclidean data has several advantages and drawbacks. Among the advantages, we may cite simplicity of computation, rapid convergence, better final distortion, no parameter to tune and deterministic results, good partition of the data into homogeneous clusters; among drawbacks, uneven visualization properties, unbalanced classes and strong dependence on the initialization. As DSOM is limited to a finite space (prototypes being chosen among samples), the dependence on the initial values is very strong. To overcome this issue, the algorithm has to be run with several different initializations.

Graphically, self-organizing maps may be represented alternatively as shown in Figure S1a and b. The first map represents a rectangular 4×4 grid (Figure S1a). Nodes and cells are always indexed in a lexicographic order from 1 for the node/cell in the upper left corner to M in the lower right corner. This graph is a faithful image of the grid structure and of distances between prototypes (i.e. nodes). However, this kind of graph is not always available as the coordinates of the prototypes in $\mathbb{R}^2$ are not generally known. What are usually known are the graph topology and distances between prototypes. In this case, the representation in Figure S1b is generally used (Cottrell & de Bodt 1996). Each cell k of Figure S1b is now associated with the corresponding node and subset A_k. Edges between nodes are suppressed, thus there is no need to know the exact coordinates of the prototypes. Instead, distances with respect to the nearest neighbours are computed for each node. The distance between two nodes/cells is computed as the dissimilarity between the corresponding prototypes. Inner cells have eight neighbours, boundary cells five neighbours and corner cells three neighbours. Then, a polygon is drawn within each cell with vertices proportional to the distances to its neighbours. If two neighbour prototypes are very close, then the corresponding vertices are very close to the edges of the two cells. If the distance between neighbour prototypes is very large, then the corresponding vertices are far apart, close to the centre of the cells.

Workflow for DNA Barcode data

Like most projection methods, NLM and SOM do not accept zeros outside the diagonal of the dissimilarity matrix. However, this systematically occurred for DNA Barcode data because specimens belonging to the same species had identical DNA Barcodes or because the presence of missing data led to zero pairwise dissimilarities. Therefore, we propose the following workflow for visualizing Barcode data.

Step 1: Data clean-up. This step was necessary because the presence of missing data blurred inappropriately computations in Step 2 below. Figures S2 and S4 show that the missing data ratios per site were quite high at the beginning and the end of sequences. All the sites containing more than a prescribed threshold were removed. After removing these sites, some specimens still had many missing sites and were temporarily removed for building the algorithm. These specimens were reintroduced afterwards.

Step 2: Haplotypes selection. The cleaned up data were collapsed into distinct sequences using FaBox (Villesen 2007). As FaBox added an additional category for missing data, the resulting data set was not the accurate set of distinct haplotypes. However, this precise knowledge was not necessary in our context. After collapsing the cleaned up data, a new data set $X' = (x_1, \ldots, x_n)$ with $n < N$ distinct sequences was obtained together with a new observed variable containing their respective frequencies $p = (p_1, \ldots, p_n)$.

Step 3: Dissimilarity matrix computations. A dissimilarity matrix $D = (d_{ij})_{i,j=1,\ldots,n}$ was computed for the n distinct sequences, where $d_{ij} = d(x_i, x_j)$ was the Kimura 2P-distance (Kimura 1980) or the p-distance (i.e. normalized number of differences between sequences) computed with the R-package 'ape' (Paradis *et al.* 2004). As the dissimilarity matrix still had some zeros outside the diagonal because of missing data, the whole dissimilarity matrix, except the diagonal, was shifted with a very small positive value ε equal to a tenth of the smallest strictly positive dissimilarity value. This led to a new matrix denoted D_ε.

Step 4: DSOM processing. The DSOM algorithm was performed for rectangular grids with M cells from the dissimilarity matrix D_ε. To give more weight to the sequences appearing more often in the original data set, the representation step was modified changing the extended within-class variance by

$$E_k(x_j) = \sum_{i=1}^{n} p_i K(h(c(x_i), k))\, d(x_i, x_j).$$

The DSOM process results in M prototypes m_k, a data partition in M subsets and a self-organizing map with M polygons displaying the distances between prototypes (Figure S1b).

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Step 5: Multidimensional scaling and HAC. The NLM algorithm was run on the *M* prototypes leading to a final 2D visualization of the Barcode data set. We also performed a hierarchical agglomerative clustering using Ward's algorithm on the dissimilarity matrix of the *M* prototypes obtained with DSOM.

As NLM is a projection method only, it does not structure data so that results become rapidly unreadable when the sample size increases. Meanwhile, the DSOM algorithm produces homogeneous partitions of data into cells, but a drawback of its BATCH version is that pairwise distances between data points are not usually correctly preserved, that the projection is not optimized and quite sensitive to initialization. Hence, the steps 4 and 5 were introduced to use both the advantages of DSOM and NLM. We used DSOM for organizing purposes only and we improved the projection by running NLM. To bypass the high dependence on initialization of the BATCH DSOM, 200 random initializations were systematically performed. The only parameter to tune is the grid structure, an *a priori* choice left to the user. Actually, the map has to be large enough to achieve nonlinear projection and generate homogeneous clusters, and small enough to reduce dimension. In general, a number of cells three to five times smaller than the number of observations provides good results.

The specimens containing too many missing data, excluded from the workflow because they were less reliable, were used *a posteriori* and projected on the self-organizing maps as supplementary data.

Nonlinear mapping and DSOM are distance-based methods. We were interested whether the choice of the distance had an impact on the results. We tested both Kimura 2P and Hamming distance on the two data sets.

The script of the workflow is a R-program given as Supporting Information Data S1.

Neighbour Joining

A classical tree-based approach of species delimitation was used to describe real data sets. As our aim was to provide a robust and rapid identification of taxa rather than an accurate determination of their phylogenetic relationships, 'a fast and simple to use' tree building method such as a phenetic (distance based) tree-generating algorithm was performed. A neighbour-joining (NJ) tree based on K2P distances (Kimura 1980) was created with PAUP 4b10 (Swofford 2000) to provide a graphical representation of the patterning of divergences among species (Saitou & Nei 1987). The tree-based criterion of reciprocal monophyly was used to define species boundaries (see Knowles & Carstens 2007 for a discussion on the limits of this criterion).

Data sets

We used the data set of *Astraptes* (an Amazonian butterfly) and a data set on *Hylomyscus* (an African rodent). These two data sets differ mainly with respect to the intraspecific variability and the population mutation rate θ. Moreover, for *Astraptes*, DNA sequences were CO1 gene sequences, while they were Cytb gene sequences for *Hylomyscus*. For small mammals, this gene is widely accepted to resolve species boundaries, but *Nicolas et al.* (2012) showed that the Cytb and CO1 genes were equally efficient for the *Praominyi* tribe, to which the genus *Hylomyscus* belongs. For the clean-up step, the two thresholds were fixed to 0.10.

Astraptes fulgerator data. The data set *Astraptes fulgerator* was previously used to demonstrate the synergy between DNA barcoding and morphological/ecological diversity studies: a combined morphological, natural history and barcoding approach revealed a complex of 10 species in one small area of Costa Rica (Hebert *et al.* 2004). The initial data contain 663 bp (sites) of the CO1 gene for 465 specimens. The specimens labelled NUMT and MYST (six specimens in all) were excluded from the analysis according to Hebert *et al.* (2004). After the clean-up step (see Figure S2), the data set contained 440 specimens and 632 sites. The 19 removed specimens mostly came from oversampled species. The population mutation parameter is $\theta = 14.32$ (Watterson estimate). The variability within species is very low, with only 1 to 22 distinct sequences per species (Table 1), with rather high interspecific variability, although there is no barcoding gap (Figure S3). The NJ tree (Fig. 1) revealed 10 groups corresponding to the 10 species which was consistent with Hebert *et al.* (2004).

Hylomyscus data. Contrary to *Astraptes*, this data set has been much less studied in the Barcoding context and its taxonomy is still debated. Wood mice of the genus *Hylomyscus* (Thomas 1926) are small-sized rodents belonging to the family Muridae (Musser 2005; Carleton *et al.* 2006). They are geographically restricted to Tropical Africa, where they are abundant in forest and dense vegetation (Rosevear 1969; Nicolas & Colyn 2003). Based on external and craniodental morphology, *Hylomyscu*s were separated into six species groups: *aeta*, *alleni*, *anselli*, *baeri*, *denniae* and *parvus* (Carleton *et al.* 2006). Within these species groups, *Hylomyscus* specimens are morphologically similar and, as a result, the taxonomy of each species is still debated. Recently, Nicolas *et al.* (2006, 2008, 2010) showed that mitochondrial DNA sequences of the Cytb gene could help solving taxonomical difficulties within this genus and could be used as a suitable molecular marker for the diagnosis of *Hylomyscus* species.

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Table 1 Number of specimens and distinct sequences within species for *Astraptes*

Species	BYTTNER	CELT	FABOV	HIHAMP	INGCUP	LOHAMP	LONCHO	SENNOV	TRIGO	YESENN
Nb specimens	3	23	31	16	65	44	38	95	51	74
Nb of distinct sequences	1	8	11	4	6	6	7	24	8	22

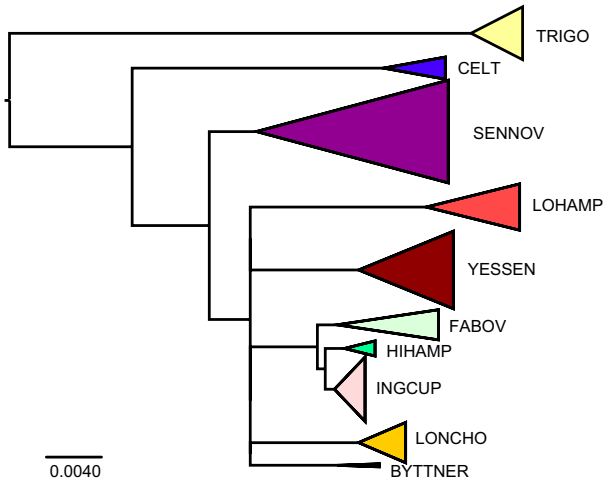


Fig. 1 NJ tree based on K2P distances of CO1 DNA sequences of *Astraptes*.

The data set consisted of Cytb sequences (1134 bp) of 454 specimens belonging to 10 widely recognized species and four taxa which probably represent undescribed species based on unpublished morphometric data. After the clean-up step (Figure S4), the data set contained 413 specimens and 731 sites. The Cytb gene was twice as long as CO1 and possessed much more missing data (especially at the beginning and in the second part of the gene). Removing the sites with more than 10% missing values reduced the number of sites, leading to a length comparable to the CO1 gene. The 41 suppressed specimens mostly came from oversampled species. The population mutation parameter is $\theta = 45.16$ (Watterson estimate). The variability within species is higher than for *Astraptes*, with 2 to 70 distinct sequences per species (Table 2), although there is no barcoding gap (Figure S5). The NJ tree revealed 16 species (Fig. 2). Twelve of the 14 included species were monophyletic in the NJ tree (*H. kaimosae*, *H. stella*, *H. walterverheyeni*, *H. alleni*, *H. simus*, *H. pamfi*, *H. grandis*, *H. baeri*, taxon1, taxon 2, taxon 3 and taxon 4). Deep divergences within *H. parvus* and *H. aeta* were observed, and these species appeared paraphyletic in the NJ tree (each species was composed of two unrelated species).

Results

The colour codes of the pie charts describing the species abundance (Fig. 3 and 7) were used for species identification in the various visualization outputs (Figs 4–6, S6 and Figs 8–10, S10). Each prototype was labelled as its corresponding cell and coloured according to the majority species in the cell. Numbers in the cells in Figs 5 and 9 indicate how many distinct sequences were clustered in each cell. The distances between cells were represented by polygons coloured according to the species. Each polygon was drawn according to the rule described in the paragraph on self-organizing maps. NLM projection provides information about proximities and distances between species, which are preserved independently on the 2D structure obtained with DSOM. Edges on the NLM projection performed on the DSOM prototypes are drawn if the corresponding cells are neighbours in the SOM representation. An alternative representation to NLM is to perform hierarchical clustering, although the relative positions of groups are no longer available. The distribution of the prototypes obtained by DSOM among the haplotypes is displayed in Figure S11. Let us note that, for both data sets, the prototypes did not systematically correspond to the most frequent sequences.

The workflow was applied without any prior knowledge on species. This *a priori* information was used after-

Table 2 Number of specimens and distinct sequences within species for *Hylomyscus* (*Hae* = *H. aeta*, *Hal* = *H. alleni*, *Hba* = *H. baeri*, *Hgr* = *H. grandis*, *Hka* = *H. kaimosae*, *Hpa* = *H. pamfi*, *Hpr* = *H. parvus*, *Hsi* = *H. simus*, *Hst* = *H. stella*, *Ht1* = taxon 1, *Ht2* = taxon 2, *Ht3* = taxon 3, *Ht4* = taxon 4, *Hwa* = *H. walterverheyeni*).

Species	*Hae*	*Hal*	*Hba*	*Hgr*	*Hka*	*Hpa*	*Hpr*	*Hsi*	*Hst*	*Ht1*	*Ht2*	*Ht3*	*Ht4*	*Hwa*
Nb specimens	14	36	2	4	11	35	14	58	71	3	27	9	26	103
Nb distinct sequences	12	24	2	3	5	21	13	42	36	2	21	7	18	70

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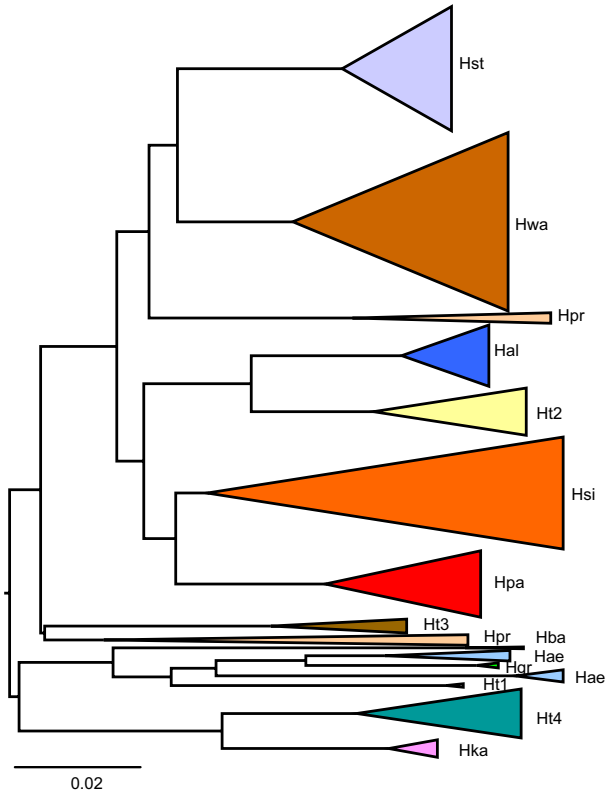


Fig. 2 NJ tree based on K2P distances from Cytb DNA sequences of *Hylomyscus* data set. Hae = *H. aeta*, Hal = *H. alleni*, Hba = *H. baeri*, Hgr = *H. grandis*, Hka = *H. kaimosae*, Hpa = *H. pamfi*, Hpr = *H. parvus*, Hsi = *H. simus*, Hst = *H. stella*, Ht1 = taxon1, Ht2 = taxon 2, Ht3 = taxon 3, Ht4 = taxon 4, Hwa = *H. walterverheyeni*.

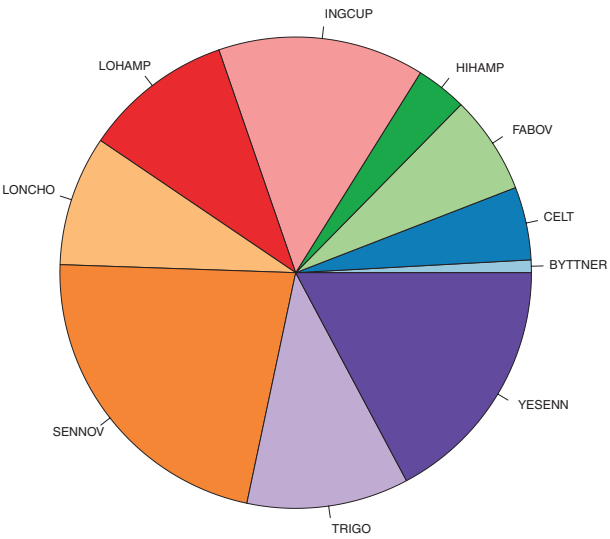


Fig. 3 Pie chart of species abundance for *Astraptes fulgerator* and species colour code used for visualization representations.

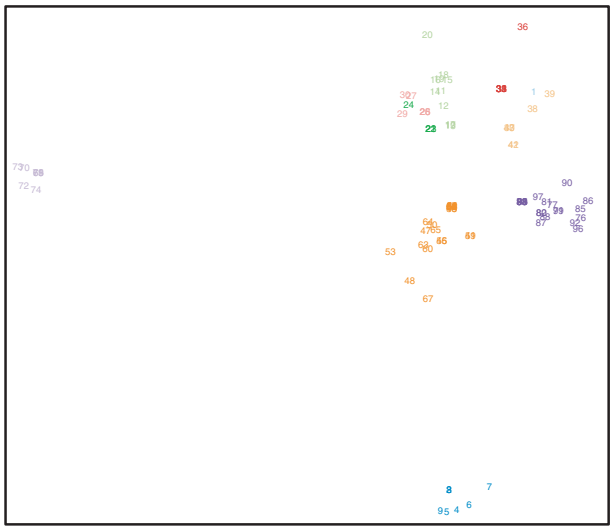


Fig. 4 Non Linear Mapping (NLM) performed on the 97 distinct sequences of *Astraptes fulgerator*.

wards to assess the results. The last step of the analysis consisted in the study of the specimens that had been suppressed from the analysis since containing too many missing values and hence unreliable. They were considered as supplementary specimens. Each specimen was projected on the map by assigning it to the cell containing its closest prototype. The quality of the projection was measured by comparing the species in the cell to the species of the projected specimen.

Since Kimura 2P and Hamming distance on the two data sets yielded similar results, only the results obtained with the K2P distances are presented here.

Astraptes fulgerator

The relative abundance of the 10 species in the original data set is displayed in Fig. 3. The various species were not equally sampled and appeared quite unbalanced. Figure S5 displays the NLM results on the 440 specimens of the cleaned up data. Three groups could be distinguished, two corresponded to CELT and TRIGO whereas the third one mixed all the other data. Figure 4 displays the results of the NLM algorithm on the 97 distinct sequences obtained with the Step 2. CELT and TRIGO were still projected into homogeneous groups far from the others, but other groups were also quite well separated. The oversampled species YESENN and SENNOV were projected into homogeneous groups while HIHAMP, INGCUP and FABOV were overlapping. Following Wang *et al.* (2010), all the specimens from the two outlying clusters, CELT and TRIGO, were then removed from the analysis and NLM was run on the remaining sequences. Results of this additional analysis are displayed in Figure S8. Except HIHAMP and FABOV, the

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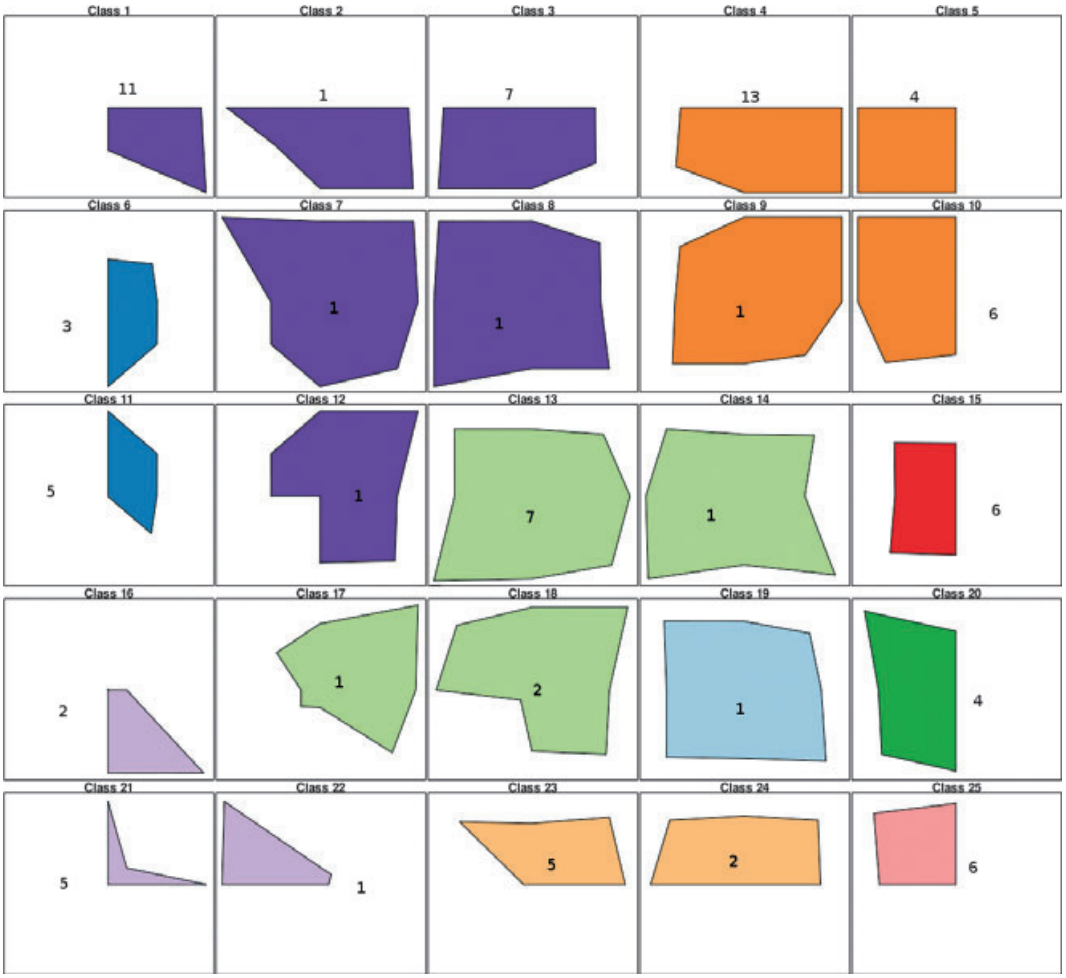


Fig. 5 Rectangular 5 × 5 grid Self-Organizing map (SOM) for *Astraptes*. Polygons within cells are coloured according to the majority species and drawn with vertices proportional to the dissimilarity between neighbour cells, close cells having polygons with close vertices.

other species corresponded to quite well separated groups. Step 3 was then applied to the K2P dissimilarity matrix with a shift of ε equal to 0.00001. For Step 4, a 5 × 5 rectangular grid was chosen which agrees with the usual choice for M (three to five times smaller than the number of distinct sequences). Figure 5 shows the self-organizing map stabilized with various initializations. No cell mixed specimens from different species. On the map, the three cells (16, 21, 22) comprising TRIGO specimens are very close together and far from others. This also holds for the two cells (6, 11) containing CELT. Step 5 was run on the prototypes. Results are displayed in Fig. 6. Figure S7 shows the result for HAC performed on the K2P-dissimilarity matrix of the 25 prototypes obtained with DSOM. Eight groups might be clearly discriminated (corresponding to TRIGO, CELT, SENNOV, YESENN, LOHAMP, BYTTNER and LONCHO specimens), while FABOV, HIHAMP and INGCUP were still mixed. But, in the additional analysis, INGCUP was no longer mixed (Figure S8). The last step consisted in the study of the 19 specimens that had been removed. They were projected on the 5 × 5 SOM and assigned the species of the nearest prototype. No inconsistency was observed: all the specimens were correctly assigned to their species.

Hylomyscus

Figure 7 describes the abundance of the 14 species in the original data set. Species were clearly unbalanced, some being highly under sampled (e.g. *Hba, Hgr, Ht1, Ht3*). The results of the NLM algorithm run on the 413 clean-up sequences are displayed in Figure S9, and those of the NLM algorithm applied on the 276 distinct sequences in Fig. 8. Both projections, performed on quite large samples sets were difficult to analyse. However, they showed that some species had very low intraspecific variability (e.g. *Ht4*, *Hst*, *Hka*, *Hba*, *Hpa*, *Hal*), whereas others

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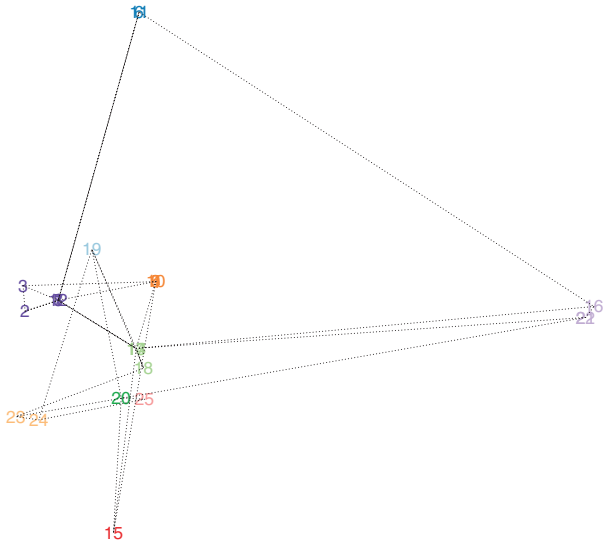


Fig. 6 Non Linear Mapping on the 25 prototypes resulting from DSOM on *Astraptes*. Edges are drawn between prototypes if they are neighbours in SOM.

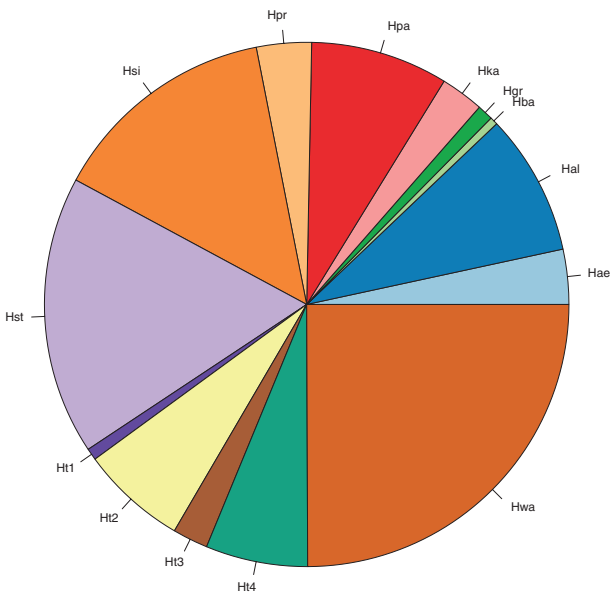


Fig. 7 Pie chart of species abundance for *Hylomyscus* data set and species colour code used in the visualization figures. Hae = *H. aeta*, Hal = *H. alleni*, Hba = *H. baeri*, Hgr = *H. grandis*, Hka = *H. kaimosae*, Hpa = *H. pamfi*, Hpr = *H. parvus*, Hsi = *H. simus*, Hst = *H. stella*, Ht1 = taxon1, Ht2 = taxon 2, Ht3 = taxon 3, Ht4 = taxon 4, Hwa = *H. walterverheyeni*.

had high intraspecific variability (e.g. *Hae*, *Hwa*, *Hsi*, *Hpr*). Contrary to the *Astraptes* data set, NLM did not show the existence of outlying clusters that could be removed to improve the resolution as in Wang *et al.* (2010). Step 3 was then applied to the K2P dissimilarity matrix with a shift ε equal to 0.00001. For Step 4, a 9 × 9 rectangular grid was chosen, and the self-organizing map is displayed in Fig. 9, together with the number of haplotypes in each cell. No cell contained specimens from different species. Except *Hgr* (three cells), the undersampled species (e.g. *Hba, Ht1, Ht3, Hka*) were projected on only one or two cells of the map. Some relatively abundant species were mapped on a low number of cells: *Hal* (cells 9, 17, 18, 27), *Hst* (cells 1, 2, 10, 11). On the contrary, *Hae, Hsi* and *Hwa* spanned, respectively, on 7, 19, 24 cells of the map, and these cells might not be close within a species. No cell contained sequences from different species. Figure 9 also displays the distances between the 81 cells with polygons coloured according to the species. NLM was then run on the dissimilarity matrix of the 81 prototypes and results are displayed in Fig. 10, with edges between neighbour SOM prototypes. The cells comprising the two species *Hal* (cells 9, 17, 18, 27) and *Hst* (1, 2, 11, 19) were very close indicating low within variability, while other species were projected in several groups (e.g. *Hpr, Hwa, Ht2, Ht4, Hae*). For *Ht2* (cells 8, 25), *Ht4* (28, 29, 37, 38, 39, 46, 47, 55, 64) and *Hpa* (10, 43), two subgroups could be identified within each species. However, the distance between them was lower than those found between species. For *Hae* (48, 49, 56, 57, 58, 65, 74), *Hpr* (3, 72) and *Hwa* (spanning on 24 cells), two to three subgroups could be identified with a distance between them higher than those generally found between species. Thus, a cryptic diversity might exist for these three species. Figure S10 displays the result of HAC on the K2P-dissimilarity matrix of the 81 prototypes obtained with DSOM. The dendrogram highlighted the existence of at least two groups within *Hae, Hwa, Hsi and Hpr*, which would lead to 18 groups instead of the 14 species. Finally, the 41 specimens removed by the clean-up step were assigned to the cell containing the closest prototype, leading to a species assignment. All the specimens were correctly assigned.

Discussion

Two distances, the Hamming distance (i.e. normalized number of differences) and the K2P distance, were considered for these analyses. The results were quite similar, and we presented here the K2P results. In a previous work concerning assignment problems for other African rodents (Nicolas *et al.* 2012), we observed that the K2P distance yielded slightly better results than the Hamming distance. Hence, we are reluctant to recommend a systematic use of the Hamming distance.

Some lessons from the Astraptes data set

Studying first a well-known data set such as *Astraptes* was very informative and provided a good starting point

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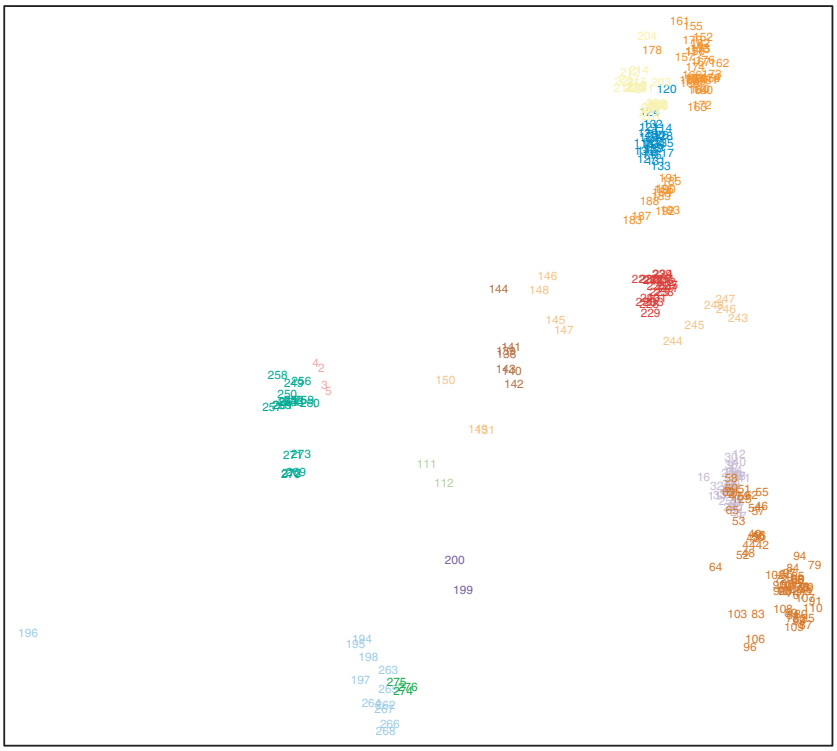


Fig. 8 Non Linear Mapping (NLM) performed on the 276 distinct sequences of *Hylomyscus*.

for designing visualization algorithms for Barcode data. This yielded a good illustration of the advantages and drawbacks of various methods in the context of Barcoding. The first natural step was to use a visualization algorithm such as Non Linear Mapping. However, NLM could not simply be applied as the algorithm did not accept 0 outside the diagonal of the dissimilarity matrix. For Euclidean data sets, dissimilarities between observations are seldom equal to 0 while this systematically occurred for Barcode data: two specimens from the same species often had dissimilarities equal to 0 because they had either identical Barcode sequences or sites with missing values, skipped in the distance computations. Therefore, the original data set was summed up into a set of distinct sequences with their respective frequencies in the original data set. Fast algorithms, as FaBox, consider missing values as a supplementary qualitative value. Hence, this additional step is sensitive to missing values. As Barcode data had often missing values, especially at both ends of sequences, a clean-up procedure was introduced consisting in removing first the sites with too many missing values and second removing the specimens still having too many missing sites. Let us stress that we did not need to find precisely the haplotypes but just to find rapidly a set of reasonably distinct sequences.

Applying NLM to the 440 *Astraptes* DNA sequences yielded rather disappointing results. Indeed, Figure S5 clearly discriminated TRIGO, CELT and LOHAMP, but all the other species were mixed up. The genetic divergence within the *Astraptes* data set was very low (97 haplotypes from the 440 CO1 sequences) and some species identified by Hebert *et al.* (2004) shared very low interspecific variability. Moreover, Figs 4 and S5 clearly showed that the two clusters TRIGO and CELT were far away from the rest of the sequences. This allowed improving the resolution by running NLM without these two species. As the number of haplotypes was reasonably small, the NLM algorithm yielded quite good results for visualization (Figs 4 and S8), the additional steps DSOM+NLM clarifying the previous representation. Species with very low intraspecific variability, TRIGO, LOHAMP and BYTTNER (1 haplotype) were very compact when projected (resp. 3, 1, 1 cells), whereas species with higher intraspecific variability, YESENN, SENNOV and FABOV, had widely spread projections (resp. 6, 4, 4). Results of the NLM, either on the haplotypes or on the DSOM prototypes (Figs 4 and 6) were very similar. This is an important remark as it points out that applying DSOM in a case where data were already well structured did not blur the results (invariance to DSOM). The projections of TRIGO and CELT were always far from the others, which could mean that they are probably very different from the others, whereas the projections of FABOV, INGCUP and HIHAMP were systematically very close or overlapping. According to

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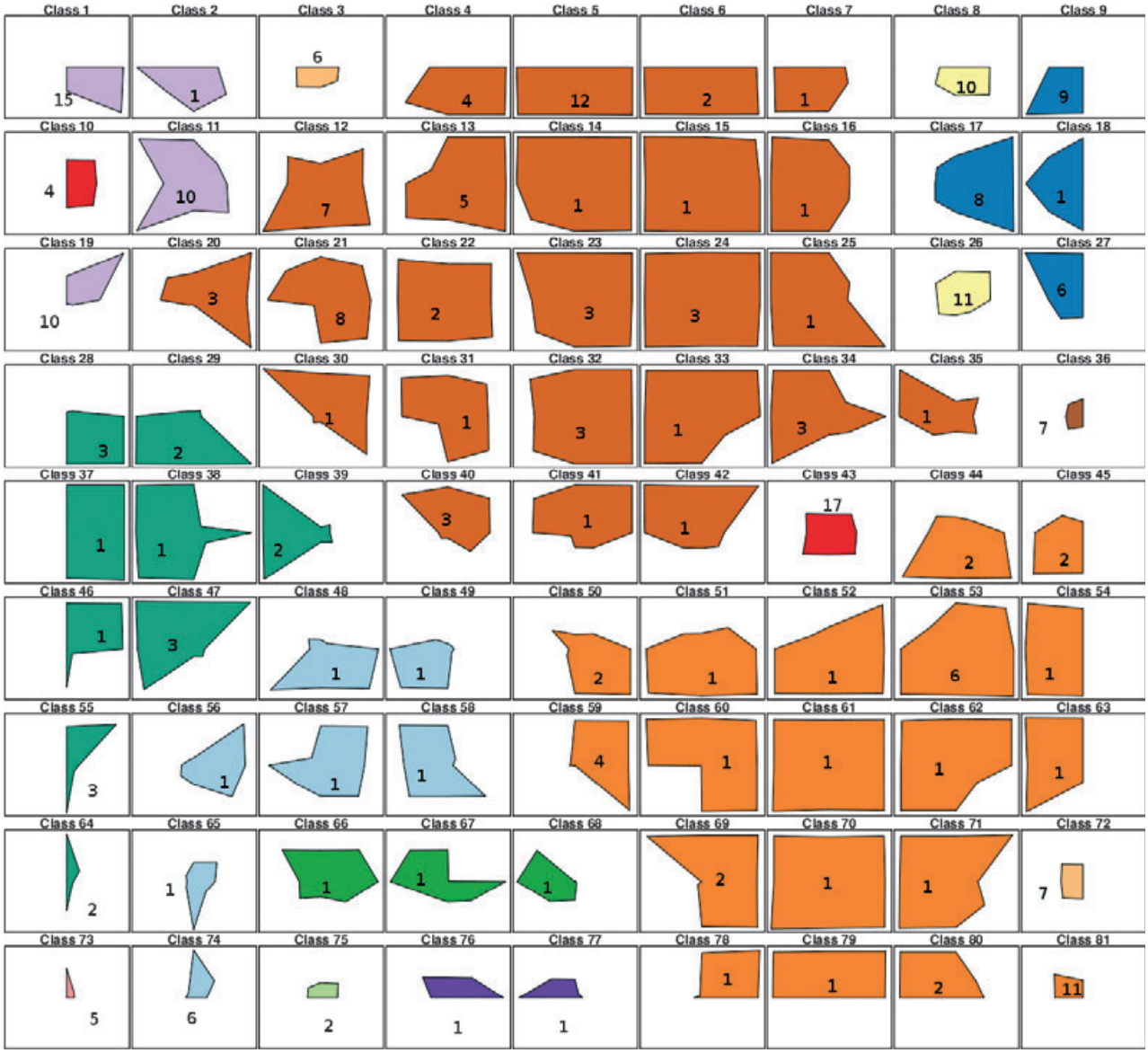


Fig. 9 Rectangular 9 × 9 grid- Self-Organizing map (SOM) for *Hylomyscus*. Polygons within cells are coloured with the majority species colour and drawn with vertices proportional to the dissimilarity between neighbour cells, close cells having polygons with close vertices.

Hebert *et al.* (2004), these three species show very little interspecific variability (less than 0.5%). As a consequence, distance-based methods cannot separate these three groups. However, for this data set, Brower (2006) was dubious about the existence of 10 species and concluded after a bootstrapped NJ analysis that there were at most six to seven species. This is consistent with the results obtained with NLM on haplotypes (Fig. 4), and NLM or HAC on the prototypes of DSOM (Figs 6 and S7) where seven or eight groups could be identified. We also observed on this data set that the DSOM-MDS methodology was overall robust to undersampling and oversampling. However, for a strongly undersampled species (e.g. BYTNNER: three specimens, one haplotype), projecting it on one specific cell was difficult. Another issue arose concerning the sensitivity of DSOM to initialization. The way to bypass these two issues consisted in performing multiple initializations of DSOM. For both data sets, 200 random initializations were sufficient, which remained reasonable in terms of computational costs as the DSOM algorithm was quite fast. Thus, our workflow is able to manage with unbalanced groups within data as long as they are not too unbalanced. This issue is very important for Barcode data because, in field

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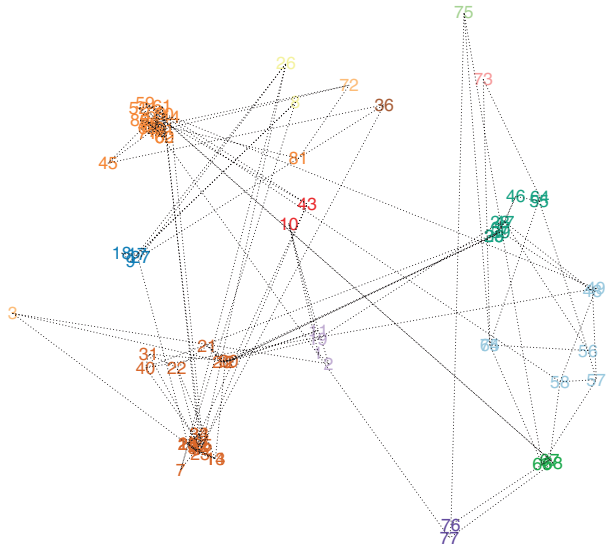


Fig. 10 Non Linear Mapping (NLM) on the 81 prototypes resulting from DSOM on *Hylomyscus*. Edges are drawn between prototypes if they are neighbours in SOM.

studies, some species are always easier to sample than others, leading to unbalanced groups.

Studying a complex data set

The *Hylomyscus* data set was clearly a much more complex data set as it is very diverse with a still debated taxonomy. Therefore, it is quite valuable to propose visualization and/or clustering tools which could be used without any a priori knowledge. As expected, the NLM visualization based on the 413 cleaned up sequences was unreadable. The NLM algorithm on the 276 haplotypes was better, but still not satisfactory. Because of the complexity of the data set, we had better choose a graph with nodes in the upper limit, given the number of haplotypes. The results of DSOM displayed in Fig. 9 showed that most species were projected on one or several cells of the map and no cell contained specimens from different species. Species with high variability, *Hwa* and *Hsi* spanned onto many cells (resp. 24 and 19). The taxa *Ht1* (2), *Hgr* (3), *Ht4* (4), *Hka* (5), *Hst* (36), *Hal* (24), *Hba* (2), *Ht3* (7) were well discriminated in both projections. *Hpr* projected by DSOM on two distant cells (3, 72) were still distant with NLM, whereas *Hpa* (cells 10, 43) and *Ht2* (cells 8, 26) had very close projections with NLM. For *Hae*, *Hpr*, *Hwa* and *Hsi*, several subgroups were observed, using either the NLM projection or HAC (Figs 10 and S10), with a distance between these clusters similar or even higher than the interspecies distance. The NJ tree displayed two groups for *Hae* which did not cluster together. These two groups within *Hae* were also identified by our method, one group comprising all the specimens from the Congo River right bank, the other one with all the left bank specimens. Two distant cells (3, 72) corresponded for *Hpr* in the SOM projection. On the NLM and HAC analyses, they were also distinct. These results suggest the existence of a discontinuity in the genetic distances distribution for these two species, which did not clearly appear with histograms (especially for *Hpr)*. Looking at the two *Hpr* subsets of the SOM partition corresponding to these two cells revealed that all the specimens from Gabon and Congo were gathered in one group, and specimens from DRC (Kisangani region) in the other group. Unpublished morphometric analyses on craniodental measurements confirmed the existence of two distinct groups in *Hae* and *Hpr*. The NJ tree displayed two groups within *Hpr* which did not cluster together. Within *Hwa*, Figs 9 and 10 identified two groups, one corresponding to cells 4-7, 13-16, 23-25, 32-34, 41, 42 (45 specimens) and the other one to cells 12, 20, 21, 22, 30, 31, 40 (25 specimens). Coming back to specimens, we observed that these two groups came from different geographic regions: the first one was only found on the Sanaga River left bank, the second one on the Sanaga River right bank. Genetic and morphometric differences between these two groups were previously underlined by Missoup *et al*. (2009). *Hsi* (42 haplotypes) spanned over 19 cells (44, 45, 50-54, 59-63, 69-71, 78-81) of the map but the polygons showed that, except for cell 81 (11 haplotypes), they were close. This was confirmed by Fig. 10, displaying the results of NLM on the prototypes, and by Figure S10 for HAC.

The projections of taxa *Ht1*, *Ht2*, *Ht3*, *Ht4* were all located on different cells of the map and did not mix with specimens of other species. *Ht4* spanning over nine cells of the map seemed to display a high within variability. However, all the cells were very close. This means that there is no discontinuity of the genetic distances between the specimens. The NLM projection in Fig. 10 also showed four well distinct groups: prototypes 76 and 77 for *Ht1*, 36 for *Ht3*, 10 and 26 for *Ht2*; 28, 29, 37-39, 46, 47, 55 and 64 for *Ht4*. This confirmed that *Ht1*, *Ht2*, *Ht3* and *Ht4* might represent four undescribed species, which is consistent with the result of the NJ tree where each taxon was monophyletic.

To conclude, Figs. 9 and 10 displayed results that were consistent with known species and could reinforce some conjectures about several cryptic species.

Advantages and drawbacks of DSOM

In general, nonlinear projection methods such as MDS or the joint algorithm DSOM+MDS are simple tools for visualizing DNA Barcode data and getting some insights about it. MDS performs projection only. Using just MDS

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on the haplotypes is well suited when the sample size is rather small. When the sample size increases, MDS meets the same issues as other unsupervised clustering techniques such as NJ: it becomes rapidly unreadable. In this case, running DSOM will reduce the dimension of the data while MDS will provide a nice mapping of the prototypes. We tried different configurations on different samples for the DSOM algorithm, all leading to the same conclusion. In general, cells were homogeneous and did not mix specimens from different species. Based on the previous results and remarks, we propose the following methodology for visualizing the DNA Barcode data:

(a) If the data set is small sized or if the number of distinct sequences is rather small, then a multidimensional scaling algorithm such as NLM could be sufficient for visualizing the data and finding groups together with their relative position.
(b) If the sample size is large, DSOM can reduce the dimension by building up a partition of the data into homogeneous clusters. Then, prototypes of DSOM are projected through a Multidimensional scaling algorithm such as NLM. The results will be more easily interpretable and the structures present in the data easier to detect.

Let us stress that both MDS and DSOM are completely unsupervised and nonparametric. This means that no *a priori* information on the species number or composition is required, and that there is no assumption about an underlying model on the data. Thus, the algorithm may be used for various data sets and in various contexts without any additional hypotheses. The unsupervised aspect of the method is interesting from two points of view. First, it allows identifying all problematic observations (poor quality, misidentification) which could lead to biased identification rules in supervised methods. Second, similarly to other unsupervised clustering methods such as NJ, DSOM+MDS can provide an insight on the groups number and composition. Moreover, the visualization properties allow assessing graphically the relative proximities or dissimilarities between groups as well as the intra- and interspecific variability. Unlike NJ trees, the MDS algorithm is not sensitive to the order of the samples in the matrix. The algorithm, like most distance-based methods was more sensitive to very low interspecific variability than to the precise choice of the distance.

Another advantage of the joint algorithm DSOM +MDS is that it is able to manage with unbalanced groups within data, which is an important issue for Barcode data as some species are oversampled and others undersampled.

Both data sets contained several specimens with an important proportion of missing values which were not considered in the analysis. They were projected *a posteriori* on the map by assigning each of them to the closest prototype. All of them were correctly projected into cells corresponding to their species. From this angle, DSOM may be used in a supervised context as a diagnostic tool for problematic specimens (because of missing data, damaged specimen, difficult assignment from morphological criteria…). The workflow presented here can be used by a taxonomist in a decision process to assign a DNA sequence to a species. This is particularly useful in the case of cryptic species. As a conclusion, the workflow proposed here could add valuable information for studying a group of species, complete other visualization methods for Barcode data (Sirovich *et al*. 2009) and also strengthen phylogenetic or supervised statistical methods.

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M.O., V.N., B.S., and C.L. wrote the paper and designed the research; M.O. analyzed the data; V.N., A.D.M., J.K., and C.D. provided the Hylomyscus sequences.

Data Accessibility

Astraptes data set: GenBank accessions AY666597–AY667060, AY7224411, AY7224412.

Hylomyscus data set: GenBank accessions JX893661 - JX893949, AY057817, JQ735483 - JQ735486, JQ735488 - JQ735502, JQ735506 - JQ735509, JQ735511 -JQ735514, JQ735516 - JQ735539, JQ735542 - JQ735543, JQ735545 - JQ735550, JQ735555- JQ735563, JQ735565 - JQ735567, JQ735573, JQ735576 - JQ735578, JQ735581 - JQ735589, JQ735591 - JQ735594, JQ735596 - JQ735601, JQ735603 - JQ735611, JQ735614 - JQ735630, JQ735632 -JQ735634.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Figure S1 Graphical representations of a SOM algorithm result. (a) prototypes are the nodes of the 4 × 4 grid; (b) each cell contains a polygon that traduces the distances to the neighbouring cells.

Figure S2 Ratio of missing data per site (a) and per sample (b) for *Astraptes* data.

Figure S3 Histograms of the intra- and interspecific K2P distances for *Astraptes*.

Figure S4 Ratio of missing data per site (a) and per sample (b) for *Hylomyscus* data.

Figure S5 Histograms of the intra- and interspecific K2P distances for *Hylomyscus*.

Figure S6 Non Linear Mapping (NLM) performed on the 440 CO1 sequences of *Astraptes*.

Figure S7 Hierarchical Clustering on the 25 prototypes from DSOM on *Astraptes*.

Figure S8 NLM for *Astraptes* without the specimens of the two "outlying " clusters TRIGO and CELT obtained in Figs 4 and S6.

Figure S9 Non Linear Mapping (NLM) performed on the 413 Cytb sequences of *Hylomyscus*.

Figure S10 HAC on the 25 prototypes from DSOM on *Hylomyscus*.

Figure S11 Prototypes distribution among the haplotypes (with their respective frequencies). Haplotypes selected as prototypes are represented in red. The first graph contains the results for the 25 prototypes selected among the 97 haplotypes of Astraptes, the second graph the 81 prototypes selected among the 276 haplotypes of Hylomyscus.

Data S1 The R program.

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